

NIDS Session Report

Start Time: 1750242464.6239555

Stop Time: 2025-06-18T16:01:41.316658

Packets Processed: 360

Anomalies Detected: 18

Packet Rate: 0 packets/sec

Protocol Stats:

total_packets: 360

tcp_packets: 243

udp_packets: 103

icmp_packets: 0

other_packets: 14

bytes_captured: 74323

start_time: 1750242464.6217744

uptime: 236.69487142562866

packets_per_second: 1.5209455018256057

bytes_per_second: 314.0034237005125

Total Alerts: 6

Alerts:

Type: ml_anomaly | Time: 2025-06-18T15:57:49.667042

Details: {'anomaly_score': -7.245962246263592, 'features': {'packet_count': 23, 'total_bytes': 1380, 'avg_packet_length': np.float64(60.0), 'std_packet_length': np.float64(0.0), 'min_packet_length': 60, 'max_packet_length': 60, 'avg_payload_size': np.float64(2.0), 'std_payload_size': np.float64(0.0), 'total_payload_size': 46, 'tcp_count': 23, 'udp_count': 0, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 1, 'tcp_ratio': 1.0, 'udp_ratio': 0.0, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 1, 'unique_dst_ports': 23, 'total_ports_accessed': 24, 'common_port_ratio': 0.0, 'high_port_ratio': 1.0, 'avg_dst_port': np.float64(50969.391304347824), 'std_dst_port': np.float64(13.886932019253656), 'min_dst_port': 50949, 'max_dst_port': 50992, 'unique_connections': 9, 'total_connection_attempts': 23, 'avg_packets_per_connection': 2.5555555555555554, 'connection_diversity': 0.391304347826087, 'max_packets_per_connection': 4, 'unique_src_ips': 9, 'unique_dst_ips': 1, 'src_ip_diversity': 0.391304347826087, 'dst_ip_diversity': 0.043478260869565216, 'private_src_ip_ratio': 0.0, 'private_dst_ip_ratio': 1.0, 'packet_rate': 5.1171708576329, 'inter_arrival_time_mean': np.float64(0.2043032321062955), 'inter_arrival_time_std':

np.float64(0.44281721677846053), 'burst_count': 16, 'burst_ratio': 0.7272727272727273, 'payload_size_mean': np.float64(2.0), 'payload_size_std': np.float64(0.0), 'zero_payload_ratio': 0.0, 'large_payload_ratio': 0.0, 'payload_size_entropy': 0, 'syn_count': 23, 'fin_count': 0, 'rst_count': 0, 'psh_count': 0, 'ack_count': 23, 'urg_count': 0, 'syn_ratio': 1.0, 'fin_ratio': 0.0, 'rst_ratio': 0.0, 'psh_ratio': 0.0, 'ack_ratio': 1.0, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T15:57:49.667029', 'packet_count': 23, 'protocol_distribution': {'tcp': 23, 'udp': 0, 'icmp': 0}, 'connection_info': {'unique_connections': 9, 'unique_src_ips': 9, 'unique_dst_ips': 1}}

Type: port_scan | Time: 2025-06-18T15:57:52.043931

Details: {'src_ip': '10.0.2.15', 'dst_ip': '75.2.76.8', 'dst_port': 443, 'protocol': 6}

Type: ddos_indicator | Time: 2025-06-18T15:58:16.090276

Details: {'src_ip': '34.233.28.173', 'dst_ip': '10.0.2.15', 'packet_size': 1494, 'protocol': 6}

Type: ml_anomaly | Time: 2025-06-18T16:01:15.224229

Details: {'anomaly_score': -7.233851790750075, 'features': {'packet_count': 100, 'total_bytes': 29411, 'avg_packet_length': np.float64(294.11), 'std_packet_length': np.float64(439.4948667504548), 'min_packet_length': 55, 'max_packet_length': 1292, 'avg_payload_size': np.float64(245.05), 'std_payload_size': np.float64(443.28301061511485), 'total_payload_size': 24505, 'tcp_count': 41, 'udp_count': 55, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 2, 'tcp_ratio': 0.4270833333333333, 'udp_ratio': 0.5729166666666666, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 8, 'unique_dst_ports': 29, 'total_ports_accessed': 29, 'common_port_ratio': 0.2708333333333333, 'high_port_ratio': 0.7291666666666666, 'avg_dst_port': np.float64(40856.3125), 'std_dst_port': np.float64(25193.434748012103), 'min_dst_port': 53, 'max_dst_port': 64469, 'unique_connections': 16, 'total_connection_attempts': 96, 'avg_packets_per_connection': 6.0, 'connection_diversity': 0.16, 'max_packets_per_connection': 16, 'unique_src_ips': 13, 'unique_dst_ips': 5, 'src_ip_diversity': 0.13, 'dst_ip_diversity': 0.05, 'private_src_ip_ratio': 0.15384615384615385, 'private_dst_ip_ratio': 0.4, 'packet_rate': 3.7794490690731033, 'inter_arrival_time_mean': np.float64(0.26726144251197276), 'inter_arrival_time_std': np.float64(0.7553994965270209), 'burst_count': 60, 'burst_ratio': 0.6060606060606061, 'payload_size_mean': np.float64(245.05), 'payload_size_std': np.float64(443.28301061511485), 'zero_payload_ratio': 0.04, 'large_payload_ratio': 0.14, 'payload_size_entropy': np.float64(1.2273038404262657), 'syn_count': 21, 'fin_count': 0, 'rst_count': 18, 'psh_count': 0, 'ack_count': 41, 'urg_count': 0, 'syn_ratio': 0.5121951219512195, 'fin_ratio': 0.0, 'rst_ratio': 0.43902439024390244, 'psh_ratio': 0.0, 'ack_ratio': 1.0, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:01:15.224211', 'packet_count': 100, 'protocol_distribution': {'tcp': 41, 'udp': 55, 'icmp': 0}, 'connection_info': {'unique_connections': 16, 'unique_src_ips': 13, 'unique_dst_ips': 5}}

Type: port_scan | Time: 2025-06-18T16:01:15.269141

Details: {'src_ip': '10.0.2.15', 'dst_ip': '3.212.203.130', 'dst_port': 443, 'protocol': 6}

Type: ddos_indicator | Time: 2025-06-18T16:01:16.034644

Details: {'src_ip': '34.198.21.75', 'dst_ip': '10.0.2.15', 'packet_size': 1494, 'protocol': 6}